THE RELATIVE FROBENIUS OUT OF A WEAKLY ETALE SCHEME OVER A CHARACTERISTIC-P BASE IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19214-19219 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19221-19228, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. First application card of the weakly etale trilogy — closes the series with a payoff used by the chapter's quasi-Frobenius discussions.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Varieties, lemma relative-frobenius (universal homeomorphism property); chapter lemmas weakly-etale-permanence (card L16090) and weakly-etale-universal-homeomorphism (card L16091).

Source reference. Original lemma 0951: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19165–19168. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F6V: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19190–19193. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CCB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19228–19228. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $U \to X$ be a weakly étale morphism of schemes where X is a scheme in characteristic p. Then the relative Frobenius $F_{U/X} : U \to U \times_{X,F_X} X$ is an isomorphism.*

Proof. The morphism $F_{U/X}$ is a universal homeomorphism by Varieties, Lemma 0CCB. The morphism $F_{U/X}$ is weakly étale as a morphism between schemes weakly étale over X by Lemma 0951. Hence $F_{U/X}$ is an isomorphism by Lemma 0F6V. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.